

Unit 7, Lesson 7: Interpreting Linear Relationships from Tables and Written Descriptions

Benchmark

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y -intercept of a two-variable linear equation from a written description, a table, a graph or an equation in slope-intercept form.

Learning Targets

- ☐ I can determine and interpret the slope and y -intercept of a two-variable linear equation from a table of values in a real-world context.
- ☐ I can determine and interpret the slope and y -intercept of a two-variable linear equation from a written description in a real-world context.

7.7.1 Warm-Up: Languages Spoken in Florida

Florida has the third largest population of all states within the U.S. While nearly $\frac{3}{4}$ of the population in Florida speaks English at home, there are actually more than 200 languages spoken in homes throughout the state. Some of these languages include Spanish, Haitian Creole, French, Portuguese, German, Tagalog, Arabic, and Mandarin.

The equation $y = 128x + 1018$ can be used to model the relationship between the number of people in Florida who speak a language other than English at home, y , in thousands, and the number of years since 1980, x .

1. Complete the statement to interpret this relationship in context.

In 1980, approximately

○ 128

○ 1,018

○ 128,000

○ 1,018,000

Floridians spoke a language other than

English at home. Since then, the number of Floridians speaking a language other

than English at home has grown by approximately

○ 128

○ 1,018

○ 128,000

○ 1,018,000

○

people every year.

7.7.2 Guided Instruction: Interpreting Slope and y -intercept from Tables and Descriptions

1. Econ Rental Car is a company that rents cars using a linear pricing plan. The company charges a daily rate in addition to maintenance fees. The total cost in dollars, y , to rent a car for x days is shown in the table.

x	y
2	97
4	187
6	277
10	457

- a. Find the slope for this relationship. Show your work.
- b. What does the slope mean in this context?
- c. Zoey noticed that the table does not include a row where $x = 0$. She said there is no way to know the y -intercept from the table. Discuss with your partner how the information in the table could be used to determine the y -intercept.

2. Carson and Jordyn determined the y -intercept using different reasoning. Their thinking is shared in the table.

Carson's Reasoning	Jordyn's Reasoning										
Carson said he noticed a pattern when looking at the table. Every time the number of days increased by 2, the cost increased by \$90. <table><tr><th>$x$</th><th>$y$</th></tr><tr><td>2</td><td>97</td></tr><tr><td>4</td><td>187</td></tr><tr><td>6</td><td>277</td></tr><tr><td>10</td><td>457</td></tr></table> The y-intercept is the value of y when $x = 0$, so he used his pattern to work backwards.	x	y	2	97	4	187	6	277	10	457	Jordyn said that she can use the rate of change to find the y-intercept. She reasoned that if the daily rate is \$45 each day to rent the car, then the rate for 2 days should be \$90 and \$180 for 4 days. The difference between the daily rate and the actual cost represented in the table must be because of the maintenance fees.
x	y										
2	97										
4	187										
6	277										
10	457										

- a. Use either student's reasoning to determine the y -intercept.
- b. What does this y -intercept represent in the context?

3. Madison planted a royal palm tree in her front yard in 2012. The tree was 12 feet tall when she planted it and has been growing at a nearly constant rate. By 2020, the royal palm measured 26 feet tall.

- a. Which two quantities are being compared in this linear relationship?
- b. Use the information to determine the slope and y -intercept in this relationship. Then, interpret each value given the context.

Feature	Value	Interpretation
rate of change		
y -intercept		

7.7.3 Guided Practice: Interpreting Slope and y -intercept from Tables and Descriptions

1. When landing, airplanes typically begin their descent at 37,000 feet above the ground and decrease altitude at a constant rate until they reach 10,000 feet above the ground. This initial descent takes about 9 minutes.

Use the information to determine the slope and y -intercept in this relationship. Then, interpret each value given the context.

Feature	Value	Interpretation
slope		
y -intercept		

2. A scuba diver exploring a coral reef begins her ascent to the surface of the water. The table models the relationship between the depth of the diver, d , in feet, and time, in minutes, t , after she begins ascending to the surface.

t	d
0	−130
2	−74
4	−18

Use the information to determine the slope and y -intercept in this relationship. Then, interpret each value given the context.

Feature	Value	Interpretation
slope		
y -intercept		

7.7.4 Your Turn!

1. Visitors to Ichetucknee Springs State Park in Fort White, Florida can rent tubes to float along the spring toward the South Takeout Point. Grayson begins his float 3.5 miles from the South Takeout Point. After 2 hours, he is 1.1 miles from the South Takeout Point.

Use the information to determine the rate of change and y -intercept in this relationship. Then, interpret each value in context.

Feature	Value	Interpretation
rate of change		
y -intercept		

2. Elijah is saving money to buy a new PlayStation. He set some money he received for his birthday aside and then added more each week. The table shows the amount of money Elijah has saved at the end of each week, m , for the number of weeks he has been saving, w .

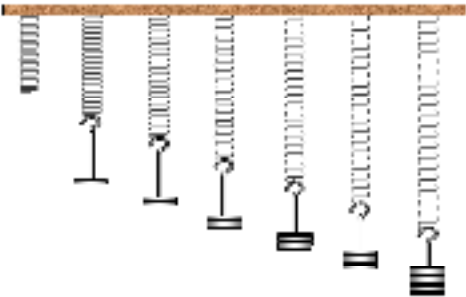
w	1	2	5	7
m	80	110	200	260

Use the information to determine the rate of change and y -intercept in this relationship. Then, interpret each value in context.

Feature	Value	Interpretation
rate of change		
y -intercept		

7.7.5 Wrap-Up: Ticket Out the Door

In science class, Aliyah observed a linear relationship between the length a spring stretches and how it changes when different weights are attached to the end. She observed the spring measures 3.7 centimeters (cm) when no weight is attached to it. When a 50 gram (g) weight is attached, the length of the spring measures 5.2 cm.



1. Use the information to determine the rate of change and y -intercept in this relationship. Then, interpret each value in context.

Feature	Value	Interpretation
rate of change		
y -intercept		